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Original Article

A Hybrid BERT-CNN Approach for Depression Detection on Social Media Using Multimodal Data

Rohit Beniwal and Pavi Saraswat*

Department of Computer Science and Engineering, Delhi Technological University, Shahbad Daulatpur, Main Bawana Road, Delhi 10042, India
*Corresponding author: pavisaraswat_2k21phdco05@dtu.ac.in

Due to the absence of early facilities, a large population is dealing with stress, anxiety, and depression issues, which may have disastrous consequences, including suicide. Past studies revealed a direct relationship between the high engagement with social media and the increasing depression rate. This research initially creates a dataset with text, emoticons and image data, and then preprocessing is performed using diverse techniques. The proposed model in the research consists of three parts: first is textual bidirectional encoder representations from transformers (BERT), which is trained on only text data and also emoticons are converted into a textual form for easy processing; second is convolutional neural network (CNN), which is trained only on image data; and the third is the combination of best-performing models, i.e. hybrid of BERT and CNN (BERT-CNN), to work on both the text and images with enhanced accuracy. The results show the best accuracy with BERT, i.e. 97% for text data; for image data, CNN has attained the highest accuracy of 89%. Finally, the hybrid approach is compared with other combinations and previous studies; it achieved the best accuracy of 99% in the categorization of users into depressive and non-depressive based on multimodal data.

Keywords: depression detection; social media; machine learning; deep learning; bidirectional encoder representations from transformers; convolutional neural network

1. INTRODUCTION

The popularity of social media platforms is increasing exponentially due to increased mobile devices and internet penetration, easy content creation and sharing facilities, growing social validation and feedback culture, etc. Various online applications, such as Instagram, LinkedIn, Facebook and Twitter, allow users to interact and communicate with each other, thereby expressing their sentiments, feelings and emotions on a particular topic [1]. Hence, these social media platforms provide an easy way for users to publicly share their opinions by responding to each other queries. However, the frequent use of social media for daily life offers significant advantages, but it involves serious drawbacks too. In this regard, state-of-the-art studies revealed that the high usage of social media platforms is directly proportional to increased depression and other mental disorder problems [2]. Good mental health is an integral part of well-being that reflects a person's psychological, emotional, social and behavioral state. On the other hand, if a person is suffering from some mental health-related issue, it may lead to serious concerns that need to be identified in the early stages.

According to a World Health Organization (WHO) report published in the year 2019, almost a billion people were facing the problem of mental disorders, out of which 14% belonged to the youth age. It is observed that teenagers are the primary source of content generators as they are primarily active on social media platforms. Furthermore, there is a significant rise in the suicide rate of youth between the ages of 15 and 29 years. As per one of the *Hindustan Times* (one of the leading news portals) reports [3], one student commits suicide every hour after posting messages like ending life or cannot survive on social media platforms. In the recent past, there have been several incidents where people have even live-streamed their suicides on social media. One of the main reasons behind this drastic decision to end life was depression [4, 5].

An estimate by WHO also revealed that mental health problems are 2443 disability-adjusted life years (DALYs) per 10 000 population and the suicide rate is 21.1, offering major economic loss in India [6]. The issues related to depression and anxiety have existed for a substantial amount of time, but their prevalence seems to have increased by 25% by the end of the first year of the COVID-19 pandemic [7]. The occurrence of COVID-19 has affected the world population to a significant extent, as a result of which their interest has shifted more toward online platforms to share feelings and thoughts. A survey by [8], as shown in Fig. 1, reveals that teenagers and young adults suffering from depression or having depressive symptoms choose to switch to social media for expressing views.

Some common mental disorders include clinical depression, bipolar disorder, anxiety disorder, post-traumatic stress disorder, autism, dementia, attention-deficit/hyperactivity disorder, obsessive-compulsive disorder and schizophrenia. Multiple reasons can cause mental illness, such as biological factors (genetics, infections, brain defects or injury, prenatal damage or substance abuse), psychological factors and environmental factors (feelings of anxiety, anger, low self-esteem or inadequacy, social expectations or dysfunctional surroundings) [9]. Among all the reasons behind the abnormal state of mind, depression is the primary cause leading to mental illness and has a significant impact on a person's mental and physical health. A depressive person can physically harm themselves or others

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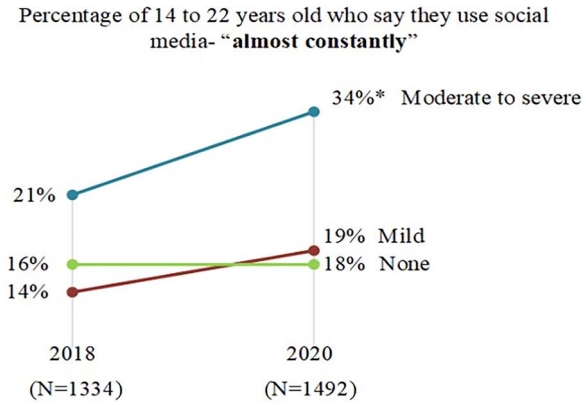

Figure 1. Frequency of social media use by depressive symptom levels (2018 and 2020) (Mahnken, 2022).

and can be the main reason for poor functioning at work or family.

Depending on the severity level, different symptoms are shown by people suffering from depression. A high severe level results in desecrating the activity of the brain and releases cortisol hormone, which has the worst impact on neuron growth in the brain. The thought process of an individual is badly affected, which can lead to suicidal attempts. A survey conducted by WHO indicates that ~280 million people are facing the issue of depression, and among them, almost 800 000 suicide cases are reported per year [10]. Due to the significant rise in depression, it has become the fourth most dangerous disease, affecting all age groups adversely.

Mental health is considered comparatively less important than physical health and is still not discussed openly because of social stigma or lack of acceptance due to understanding and awareness about it. Because of the ignorance of symptoms of mental disorders, it goes undetected for a prolonged duration, which further intensifies and leads to mental distress. Therefore, the raw data posted on social media sites by a pool of users can be helpful for researchers and doctors to critically analyze and understand a person's mental state to diagnose depression at its early stage.

The clinical evaluation of depression uses subjective techniques such as clinical interviews, rating scales and scores that may cause erroneous predictions if the observer is not an expert in the field. It is also possible that a person may convey a misleading response due to fear of unfavorable societal consequences that can further delay depression detection [11]. Moreover, the treatment of depression through medications and therapies consumes a significant amount of time and is often costly and unreliable. These traditional assessment methods require data related to numerous patients along with their history to detect depression symptoms. Also, to treat the symptoms of depression, the procedure requires continuous monitoring of the patient's day-to-day life activities to predict whether or not they have depression. Persons suffering from depression often have different behavioral symptoms in comparison to an average person, such as facial expressions, body actions, perfect speech issues, mental behavior and psychological signals, which may cause difficulty for psychologists to detect the severity level [12] accurately. Therefore, an automated and quantitative approach is required to accurately analyze and detect depression symptoms to improve a person's life span.

The advanced learning algorithms or models of artificial intelligence (AI), such as machine learning (ML), deep learning (DL), ensemble learning and transfer learning, have the remarkable ability to understand complex patterns in data and to provide accurate results with high performance. ML techniques have proven beneficial in healthcare due to their extraordinary capability of processing a large amount of data [13]. Also, natural language processing (NLP) techniques can be efficiently utilized in sentiment analysis, allowing machines to understand words and text like humans accurately. Therefore, to analyze the severity of depression of a person based on their social media posts, NLP techniques are preferred, as shown by various studies [14]. Furthermore, DL algorithms have automated the entire diagnostic process as they can efficiently deal with complex data without the intervention of human beings. These techniques allow the users to adjust the network parameters to enhance the prediction's accuracy [15]. Therefore, the incredible benefits of these computational methods have attracted the interest of researchers to utilize them for the practical analysis of persons with mental illness and depression issues.

A study by Choudhury et al. [16] used a support vector machine (SVM) to analyze the tweets of depressed users for their risk analysis and achieved an accuracy of 70%. The implication of ML and NLP algorithms can be seen in a work by Guntuku et al. [17]. They gathered tweets of the users, including user activity, genders and age, and efficiently evaluated their depression state. Similarly, Uddin et al. [18], in a research work, utilized long short-term memory (LSTM) and deep recurrent networks and accurately predicted depressive patients. Hence, due to the incredible benefits of these computational methods, they have attracted the researcher's interest to utilize them to effectively analyze persons with mental illness and depression issues.

1.1. Novelty/contributions of the study

From the survey of literature, it is found that the researchers do not significantly exploit the depression detection work based on image data. Moreover, the combination of text and image modality, i.e. multimodal, is not yet evaluated, which may yield higher accuracy and robust diagnosis. Furthermore, on multimodal data, no dataset seemed to be publicly available, which poses the foremost hurdle to carrying out this research work. Therefore, to tackle the issues and cover the undiscovered gaps in the literature, a novel hybrid framework is proposed utilizing the concept of transfer learning (TL) and deep learning (DL) based on multimodal social media data for depression detection. The significant contributions of this research work are as mentioned below:

1) As per the survey, no multimodal dataset exists for depression analysis. Therefore, firstly, a multimodal dataset is created and presented that contains the image and the corresponding text and emoticon data belonging to depressive posts.
2) Secondly, we propose a unified, robust and hybrid framework that can work on any type of social media platform and data for depression detection using images as well as text by extracting more relevant information for a reliable diagnosis.
3) Thirdly, a hybrid TL- and DL-based approach combining bidirectional encoder representations from transformers (BERT) and convolutional neural network (CNN) (BERT-CNN) is created to accurately classify posts as depressive and non-depressive by analyzing textual and image data separately.
4) Fourthly and lastly, we compared the proposed approach with existing state-of-the-art techniques along with other hybrid DL + ML approaches using various performance metrics. Moreover, we found that the proposed hybrid approach

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outperformed the other abovementioned approaches by a significant margin.

The remaining paper is organized as follows: Section 2 discusses the related work; Section 3 provides a detailed view of the proposed methodology, explaining each step; Section 4 expounds on the implementation and result details; lastly, Section 5 concludes the research paper and provides direction for future work.

2. LITERATURE REVIEW

Automated pattern recognition approaches can deal with complex data and can perform accurate pattern analysis with high accuracy. Different researchers adopt several such techniques to perform the detection of persons suffering from depression, a summary of which is provided in Table 1. The analysis of the literature revealed that for the classification of depressive and non-depressive posts, two paradigms are frequently adopted, i.e. ML and DL. This section presents an insight into the state-of-the-art work done considering both approaches for depression analysis using different social media platforms.

2.1. Based on ML approach

The ML process is the part of AI that involves using data and statistical models to work intelligently as a human brain to perform classification and object recognition. Various researchers utilized such approaches, including several supervised and unsupervised algorithms, to analyze the considered data. In supervised learning, the output for the given input label is already known. In contrast, unsupervised ML involves the training of models utilizing unlabeled data. Islam et al. [1] proposed a study to perform an analysis on depression detection based on online available Facebook data. They evaluated four supervised ML classifiers, namely decision trees (DT), K-nearest neighbor (KNN), SVM and an ensemble model on text data. The results showed the highest accuracy with DT, i.e. 71.2%, thus proving the efficacy of these algorithms in depression analysis. In another study by Ruz et al. [23], the issue regarding the evaluation of sentiments during unfavorable events such as natural disasters was addressed by analyzing tags, emoticons, and text data. They used different ML techniques, including naïve Bayes (NB), random forest (RF), SVM, transductive adversarial network (TAN) and Bayes factor (BF) TAN on two Twitter datasets, i.e. the 2010 Chilean earthquake and the 2017 Catalan independence referendum. The results attained revealed the highest performance, with SVM having an accuracy of 81.2%.

Reece et al. [30] employed ML techniques to detect depression markers using an image feature set collected from Instagram posts. They implemented Bayesian logistic regression (LR) and RF for strength and classification analysis. Their approach showed significant results, outperforming other methods with precision of 64%, recall of 67.7%, and F1-score of 64%. Arora et al. [32] applied SVM and Multinomial Naïve Bayes (MNB) on Twitter data to analyze the sentiments and depression of users from their Twitter posts. The study achieved the accuracies of 79% and 78%, respectively. Their study revealed the significant importance of ML algorithms in depression analysis.

An ensemble model with a majority voting platform was adopted by Kumar et al. [33] to classify depression patients as anxious and non-anxious users using Twitter data. Tweets of the users were used, and different linguistic, semantic and activity features were extracted. Their model is then compared with other ML models based on accuracy, such as MNB, RF and gradient boosting. The outcome demonstrated the highest performance using the proposed method with an accuracy of 85.09%. Gaikwad et al. [34] proposed a new approach known as the 'impact factor' in their study, where a value is assigned to each word in the text of the lexicon dataset. They applied different ML algorithms, including NB, KNN and SVM, to analyze the robustness of the proposed technique. The results indicated that the highest performance was achieved using SVM with an accuracy of 90% compared to NB and KNN with an accuracy of 64% and 73%.

Jayakrishnan et al. [35] performed multi-class emotion analysis to identify the hidden emotions from text data. They employed an SVM classifier to classify users' emotions into different categories, such as anger, happiness, fear, sad or normal. Their approach with syntactic features seemed to be effective in such analysis with an accuracy of 90%. Deshpande et al. [36] conducted a study to analyze depression symptoms by collecting Twitter data and using ML models. They employed MNB and SVM for this purpose, and the results revealed the relevance of these models in depression classification with an accuracy of 83% and 79%, respectively.

2.2. Based on DL approach

Apart from ML, DL models have proven efficacy in classification tasks with high accuracy as they allow users to make adjustments in the network by deviating their parameters. DL network is similar to ML but does not require any human expertise and intervention. The problems that remain unsolved due to the high complexity of data can be easily handled with the help of DL approach. Working toward this direction, Han et al. [24] proposed an efficient multi-tasking sentiment analysis model based on double bidirectional rated recurrent units (BiGRU). They gathered the data, including tags, emoticons and text from SentiDrugs. The efficiency of the proposed technique is then compared with other models such as LSTM, BiGRU, Memnet and Cabasse. The outcomes achieved showed the highest accuracy with their proposed model, i.e. 78.6%.

Liu et al. [27] performed a text data classification method related to medicine to analyze the consumer's health using the Medical Social Media Text Classification (MSMTC) technique. This technique achieved the highest accuracy of 87.65% and overcame the poor usage of consumer health terminology in medical text classification. Wu et al. [22] performed a study for the early detection of depression by simultaneously using textual, behavioral and environmental features type of data. They proposed an automatic DL-oriented method to detect depression with different data sources, i.e. from Heterogeneous Data Sources (D3-HDS). Their methodology yielded precision, recall and F1-score of 83.3%, 71.4% and 76.9% revealing its significance in the early detection of mental illness.

Katchapakirin et al. [29] developed an NLP-based framework for depression analysis for the Thai language using Facebook data. They employed LSTM and achieved an accuracy of 85%. Moreover, they concluded that analyzing Facebook users' behaviors using the DL framework can serve as an important tool for identifying depression levels. Lin et al. [31] applied the CNN model to Twitter data that included visual and textual features and attained an accuracy of 88.4%.

2.3. Based on combined approaches

Despite using just a single approach for the classification of depression, some researchers utilized both ML and DL platforms together and evaluated the efficacy against each other. The purpose was to determine the most effective approach to be robustly

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Table 1. Summary of literature work.

Author (year)	Study purpose	Modality	Dataset used	Classification technique	Performance	Findings
Amanat et al. [19] (2022)	To present a DL-based system to predict depression at early stages	Text	Tweets-scraped dataset (Kaggle)	RNN-LSTM (proposed), CNN, SVM, NB, DT	Accuracy: CNN = 91% SVM = 97.21% NB = 97.31% DT = 82% RNN-LSTM = 99.6%	• The proposed RNN-LSTM method attained the highest results with low error rate • In future, work can be performed by applying hybrid RNN
Gupta et al. [20] (2022)	To analyze different classification methods for both balanced and unbalanced data to detect depression	Text	Sentiment_140 and Senti-ment_tweets3 (Kaggle, Twitter)	DT, KNN, SVM, LR, LSTM	Highest results with LSTM: Accuracy = 83% Precision = 84% Recall = 75%	• The highest results using DL-based LSTM model revealed its significance in depression analysis • Future work can be extended using hybrid approach for more accuracy
Wang et al. [21] (2020)	To analyze sentiments of people on Weibo using BERT model	Text, tags, emoticons	Weibo	Fine-tuned BERT (proposed), SVM, NB, LR, CNN, LSTM	Accuracy: Proposed BERT = 75.6% SVM = 70.6% NB = 66.9% LR = 70% LSTM = 57.7% CNN = 71.1%	• The proposed fine-tuned model can be efficiently utilized in sentiment analysis • In future, study can be extended by using another potential social media platforms
Wu et al. [22] (2020)	To diagnose depression based on different type of data features simultaneously	Text, behavior and envi-ronmental features	CES-D and Facebook	D3-HDS (Deep Learning-based Depression Detection for Heterogeneous Data Sources), Word2vec + LSTM	Precision = 83.3% Recall = 71.4% F1-score = 76.9%	• The results revealed its significance in early detection of mental illness
Ruz et al. [23] (2020)	To tackle the issue of sentiment evaluation during serious circumstances such as natural disaster	Text, tags, emoticons	Two Twitter datasets: 2010 Chilean earthquake and the 2017 Catalan independence referendum	NB, SVM, RF, TAN, BF TAN	Accuracy: NB = 74.2% SVM = 81.2% RF = 72.5% TAN = 72.1% BF TAN = 76.4%	• SVM model can be robustly utilized for depression analysis in unfavorable conditions • Further work can be done considering more training samples
Han et al. [24] (2020)	To propose a multi-tasking sentiment analysis model based on double BiGRU	Text, tags, emoticons	SentiDrugs	LSTM, BiGRU, TD-LSTM, ATAE-LSTM and other models	Highest accuracy with the proposed PM-DBiGRU model = 78.6%	• The efficacy of drug level aspect-reviews can be enhanced using the proposed approach
Mustafa et al. [25] (2020)	To build a multi-classifier to classify depression tweets using Twitter data	Text	Twitter	NN, RF, 1-DCNN, SVM	Highest accuracy with proposed = 91%	• The use of a relevant and best combination of features can improve the diagnostic accuracy • Future work can be directed toward using more number of modalities
Ahmad et al. [26] (2019)	To propose a terrorism-related DL system that can classify the sentiments into extremist and non-extremist	Text, tags, emoticons	Twitter	KNN, RF, NB, SVM and DL methods (CNN, LSTM, CNN+ LSTM)	Accuracy: KNN = 72% RF = 82% NB = 71% SVM = 79% CNN = 83.09% LSTM = 85.07% CNN + LSTM = 92.06%	• The combination of proposed model attained highest results for prediction by storing current as well as past info • Further work can be directed toward considering advanced methods and contextual-aware parameters too

(continued)

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Table 1. Continued.

Author (year)	Study purpose	Modality	Dataset used	Classification technique	Performance	Findings
Liu et al. (2019) [27]	To propose a text data classification method to access health of the consumer	Text, tags, emoticons	Patient description	CNN, LSTM, Bi-LSTM, NVB, XGBoost, MSMTC	Accuracy: CNN = 86.2% LSTM = 85.7% Bi-LSTM = 86.5% NVB = 83.84% XGBoost = 77.86% MSMTC = 87.65%	• The proposed method overcomes the issue of less usage of consumer health terminology in medical text classification • In future, study can be extended by using the data for disease severity analysis also
Islam et al. [1] (2018)	To diagnose depression by utilizing a robust ML method	Text	Facebook	DT, KNN, SVM, Ensemble	Accuracy: DT = 71.2% SVM = 71% KNN = 60% Ensemble = 64%	• The employed ML models can enhance the accuracy for depression detection with low error rate • More dataset needs to be used in future to cross verify the results
Chen et al. [28] (2018)	To propose a DL-based model to capture the relevant features for multi-class depression detection	Text, tags, emoticons	Twitter	SVM, CNN, LSTM, CNN-LSTM, RNN-CNN-LSTM (proposed)	Accuracy (proposed) = 78.42%	• The designed model for multi-class sentiment analysis yielded high results showing its efficacy than existing techniques
Katchapakirin et al. [29] (2018)	To develop a NLP-based framework for depression analysis using Thai language	Text, tags, emoticons	Facebook	LSTM	Accuracy = 85%	• Analyzing Facebook users' behaviors using DL platform can serve as an important tool to identify depression levels
Reece et al. [30] (2017)	To determine the depression markers using ML algorithms	Image feature set	Instagram + CES-D	Bayesian logistic regression, RF	Precision = 64% Recall = 67.7% F1-score = 64%	• The proposed approach shown significant results outperforming other methods • In future, confidence need to be developed in subjects for their data security to gather more information
Lin et al. [31] (2014)	To diagnose the individuals with psychological stress by proposing a DL model	Text, image and behavioral	Sina Weibo, Tencent Weibo and Twitter	SVM, RF, NB, DNN (proposed)	Accuracy: SVM = 75.6% RF = 76.7% NB = 68.1% DNN = 78.5%	• The proposed DL-based DNN model can be helpful in detecting early stress and illness in depression patients efficiently

adopted in depression analysis. Chen et al. [28] proposed a DL-based model for multi-class classification to capture the relevant features for depression detection. They evaluated the proposed model, i.e. RNN-CNN-LSTM, against SVM, CNN, LSTM and CNN-LSTM on Twitter data. Their approach achieved an accuracy of 78.42%, the highest among all, thus showing its efficacy over the techniques mentioned above.

Amanat et al. [19] presented a robust system to predict depression at early stages using DL models. They used Kaggle dataset as input to the models, including CNN, SVM, NB, DT and RNN-LSTM. The results showed that the proposed RNN-LSTM method attained an accuracy of 99.6% with a low error rate compared to accuracies of CNN, SVM, NB and DT, having 91%, 97.21%, 97.31% and 82%, respectively. Ahmad et al. [26] proposed a terrorism-related DL-based system that can analyze and classify the sentiments into extremist and non-extremist. They evaluated different ML and DL methods, such as KNN, RF, SVM, NB, CNN, and LSTM, against the proposed CNN and LSTM. Their proposed model achieved the highest results with an accuracy of 92.06%.

Gupta et al. [20] analyzed the performance of different classification methods for both balanced and unbalanced data for depression detection. The text data are obtained from Kaggle and Twitter. The proposed LSTM model revealed higher accuracy than ML techniques, including DT, KNN, SVM and LR. Similarly, Mustafa et al. [25] applied NN, RF, 1-DCNN and SVM to build a multi-classifier that can efficiently classify depression tweets by analyzing data on Twitter. Among all the classifiers, the proposed

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DL model 1-DCNN has shown the highest results with an accuracy of 91%.

However, in the related past works, researchers use various classification approaches and modalities to enable depression diagnosis; the literature analysis indicated some areas yet to be explored to better detect depression patients. Therefore, an efficient strategy is proposed whose unique feature is that it applies to all types of social media platforms, like Twitter, Facebook and Instagram, because it includes multiple modalities such as text, images and emoticons. The state-of-art revealed that DL models performed better than ML models; therefore, this work proposes a DL-based hybrid BERT-CNN approach that uses multimodal social media data for depression detection. Finally, the performance of the presented approach is compared with existing state-of-the-art techniques along with other hybrid DL + ML approaches.

3. PROPOSED METHODOLOGY

This article proposed a novel framework that can work on multiple platforms and can classify persons with depression and non-depression based on multimodal data. The proposed methodology implemented in this research work is provided in Fig. 2 and consists of different phases such as dataset collection, including text, emoticons and images; preprocessing; classification; and performance evaluation. A detailed description of each phase is given below:

3.1. Dataset

A relevant dataset plays a crucial role in the depression detection process. The exploration of the literature revealed no public availability of a reliable multimodal dataset based on depression. Therefore, the significant contribution of this work is to create such a dataset and propose a novel DL hybrid approach to analyze depression from multimodal content, including text, emoticons and images. One of the most popular social media platforms, i.e. Instagram, has been used to gather the entire data. Those posts from Instagram users were downloaded and accessed, which were publicly available. The data are stored in CSV format consisting of the post, path of the image and happiness index (HI) where HI = 0 represents depressive post and HI = 1 denotes non-depressive post. The entire data are annotated as depressive or non-depressive with the help of an expert psychiatrist from Futela Hospital. The collected data were labeled into 0 or 1 after in-depth analysis and evaluation by the expert. A total of $n = 10\,295$ posts, including text, emoticons and images, were collected, out of which $n = 3431$ posts were depressive and $n = 6863$ were non-depressive, as shown in the following Fig. 3. The validation of data is performed by the health experts categorizing posts into depressive and non-depressive. Table 2 shows a sample list of data (text and images) used in this research work to classify depressed and non-depression individuals. In the table, the light gray shaded rows represent the text and emoticons data and their respective image data are presented in the rows shaded in dark gray color.

3.2. Preprocessing

Text plays a vital role and contributes majorly to comprising depression-related data as most of the users express their emotions via tweets, quotes, statuses, etc., frequently on social media. People often use text with emoticons to convey their feelings and sentiments that need to be correctly analyzed to perform accurate predictions regarding depression. Therefore, to preprocess the text, tokenization is done in which the given text is split into small units referred to as tokens [37]. Single words, entire sentences, phrases, etc. can be part of tokens. The process of tokenization involves the removal of characters such as punctuation, special characters and spacing.

3.2.1. *Text processing techniques*

Several such techniques are employed in this paper to clean and preprocess the text comprising the following [38,39]:

- **Removing Stopwords:** Stopwords refer to the most commonly used words in the Stoplist that need to be refined or filtered as they do not have much significance. Since Stopwords are not of much importance, they are removed utilizing the standard list of Stopwords.
- **Removing special characters:** Some special characters have drawbacks in the way they are utilized. Furthermore, the use of special characters can introduce unwanted noise in the data that can drastically affect the classification capability of a model. Therefore, apart from characters from A to Z and a to z, all other ASCII characters are eliminated from the data.
- **Discarding numbers:** To represent a mood, emotions, thoughts, etc., of a person, numbers cannot be utilized and should be discarded in order to have pure text characters. The presence of irrelevant data in the form of numbers can result in a high misclassification rate and decreased probability of accurate detection. Thus, the numbers are dropped from the text due to their negligible importance.
- **Removing punctuations, spaces and links:** While performing training of data, the use of punctuation can add irrelevant noise and lead to ambiguity. Each text can be treated as equal once the punctuation removal process is done. It enables the process to consume less memory and visualize the data. Also, the low-quality links are eliminated which were not contributing anything to the text information.
- **Removing repeated characters and spell righting:** Duplicated characters offer ambiguity in data and the short text abbreviations can lead to incorrect classifications. Therefore, the characters that were repetitively present in the text and are not part of the English dictionary are removed. The short forms misguiding the correct meaning of the words are also discarded. Furthermore, the words that have the possibility of being miswritten are replaced by their corrections. Spell correction is applied to the words using correct spelling suggestions based on Wikipedia, Oxford, dictionaries, etc.
- **Conversion to lower case:** In many scenarios, it is suggested to convert the entire characters into lower case using an appropriate function on each word. Therefore, the idea is applied to convert the input text into a similar casing format using the lower () method. Once this entire text preprocessing was done, we moved ahead with the emoticons preprocessing.

3.2.2. *Emoticon preprocessing*

Sometimes, people use many emoticons to express their moods, mental states and feelings. This modality also forms an integral part from which a user can be classified into depressive and non-depressive based on their posted emoticons. Therefore, the emoticons are converted into text to extract relevant details and provide useful insights about the users. For example, emoticon such as '●' is converted into 'black heart' text. The scores from emoticons can be utilized to analyze the mental state and emotions of a user. The value 0 is used to denote depression and 1 represents non-depressive post. Once the emoticons are transformed into the text, these are appended with the tokenized text due to their similar format.

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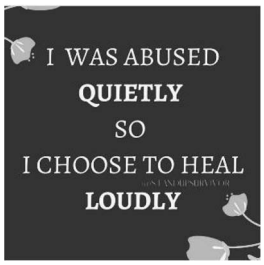
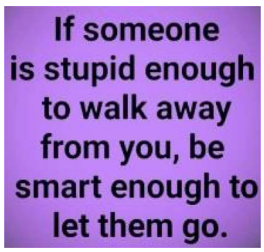
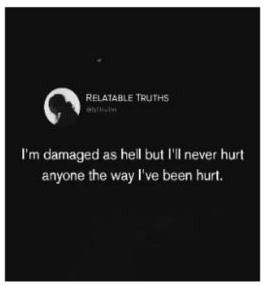
Table 2. Examples showing depressive and non-depressive posts using multimodal data (text with corresponding images).

Sample examples		
Text and emoticons	**Image**	**Category**
A lot of people do not understand what it is like to be with a narcissist. SPEAK UP AND BE LOUD! You can help others. #narcissist #narcissim #narc #narcissticabuserecovery #narcissisticabuse #narcissistawareness #narcissistic #narcissistfree #narcissistquotes #depressed		Depressive
ABSOLUTELY... 🖖#bye #narcissist #narcissisticabuse #narcissism #whytolive		Depressive
It was always all about you. Selfish piece of 💩. 🖖#bye #narcissist #narcissim #sadforlife		Depressive
But those nights feels so heavy to sleep #sleepless #sad #broken		Depressive
Let them go 🖖... #byefelicia #nobigloss #dontgobacktowhatbrokeyou before #seeyoulateralligator #dontletthedoorhitya wherethegoodlord splitya		Depressive
Relatable 🖤? #lyftruths #sadquotespage #sadquotespage #sadedits ● #sadedits #brokenquotes #sadlines #sadfeeling #sadline #lovefeelings #sadlove #brokenlove #hatelove #poetryworld #poemsbyme #depressionhelp #writer #depressededits 😷🔫 #depression #relatablequotes #relatable		Depressive

(continued)

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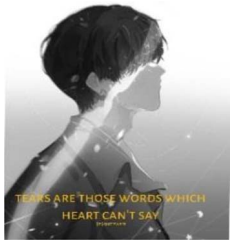
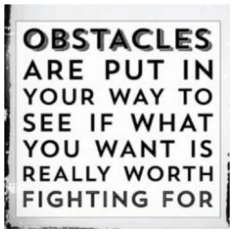
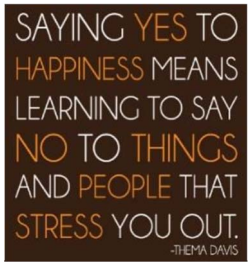
Table 2. Continued.

Sample examples		
Text and emoticons	**Image**	**Category**
Harsh Truth.. #broken #hate #depression #Nomorelife		Depressive
Fight for your dreams and never give up! #fighter #love #life #quote #obstacles #happiness #dreams #believe #selflove		Non-depressive
Enjoy your weekend and have fun! #happy #smile #fun #instahappy #funtimes #feelgood #enjoy #lovelife #laugh #weekend #selflove		Non-depressive
Absolutely love this quote!! #ceo #yourlife #happiness #freedom #picoftheday #instamood #gratitude #quote #selflove		Non-depressive
Your Past is Just a Story #moveon #goforward #letgo #past #history #live #fully #present #moment #future #destiny #create #story #life #unfiltered #truth #new #quote #selflove		Non-Depressive
Say Yes to Happiness #good #people #open #heart #give #take #learn #boundaries #yes #no #stress #relax #staycalm #letgo #accept #happiness #inner #peace #truth #mantra #new #quote #life #unfiltered #selflove		Non-Depressive

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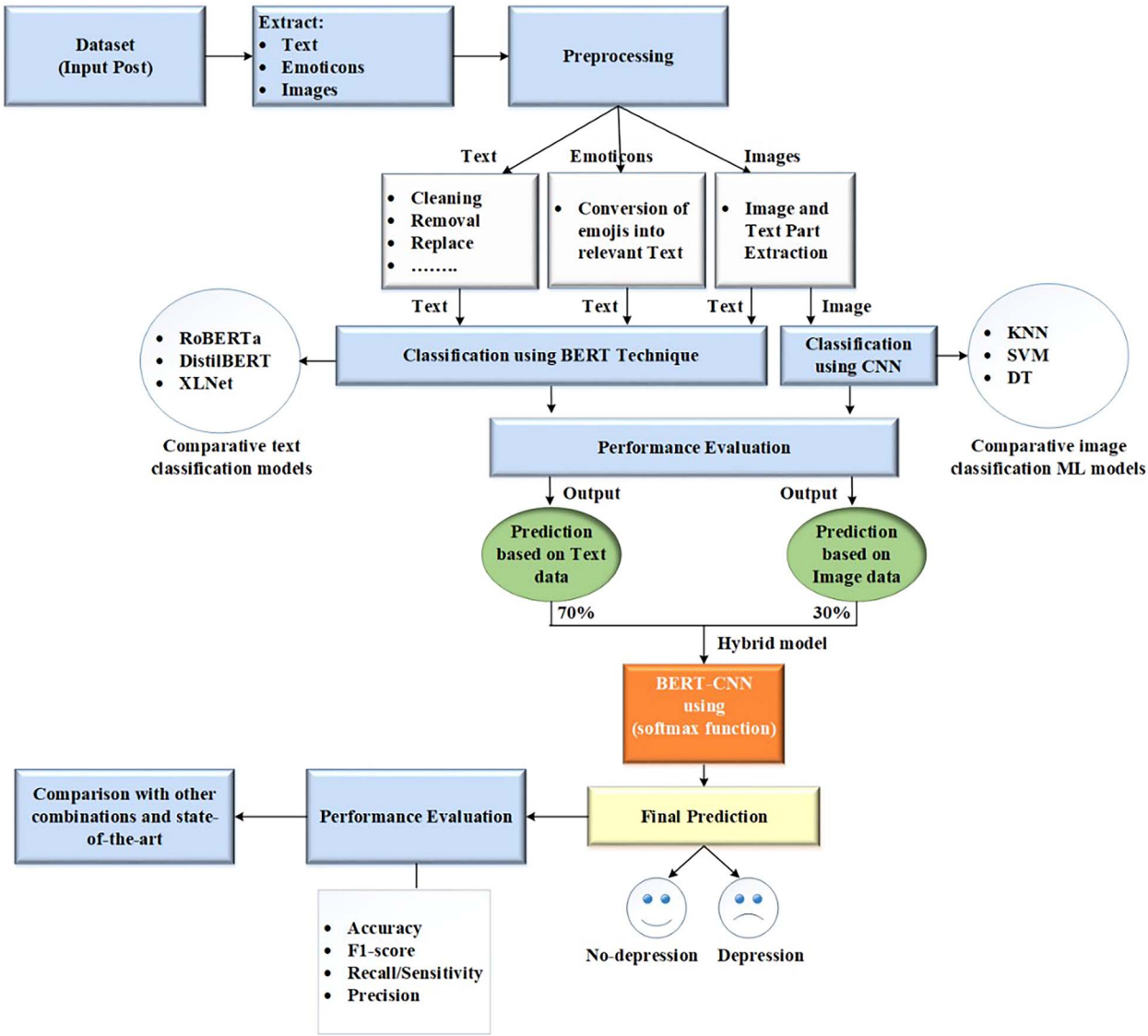

Figure 2. Proposed methodology for depression detection.

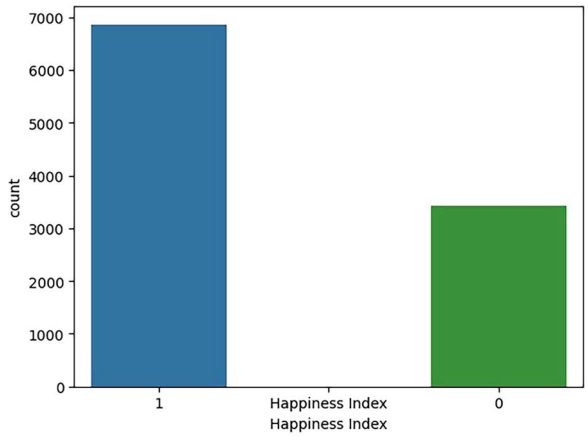

Figure 3. Dataset representations of non-depressive and depressive posts.

3.2.3. *Image preprocessing*

Like text and emoticons, image is also the most preferred modality, which, on proper visualization, can provide direct clues regarding the mental status of a person. The collected data consist of numerous images that also have the text data. Therefore, for efficient evaluation, extracting or separating the text part from the image is necessary. This extraction is performed using an 'Optical Character Recognition' (OCR) [40] technique, which has the feature to transform an image having printed or handwritten text. After extracting the text from all the images, it is processed following the same operations applied to tokenized textual data, and the image part is further processed using a DL technique.

3.3. Feature extraction

Once preprocessing is done, the data are converted into the required format to perform further operations on it. The BERT model is applied to obtain the features from text [41] to efficiently use them for the classification of a user's post. The embedding tokens of sentences are fed as input to BERT, which is followed by sentence separation using special tokens. At last, the outcome of embedding for each token, known as the hidden state, is given as output by the BERT model. Similarly, to extract the features from images, CNN is employed that does not require heavy preprocessing and extracts the features automatically without any human intervention.

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3.4. Classification

Classification refers to an ML procedure mainly supervised and used to categorize a given dataset into different classes depending on the training dataset. A sample known as a training example is provided as input to the model from which it learns and can classify the test example, which is an unknown data point. This section describes the entire process of classification using the proposed hybrid approach.

3.4.1. *Proposed approach*

The current study intends to perform classification by proposing a robust hybrid approach that can work on multimodal data. Therefore, to achieve the purpose of classification, the text and image parts are processed separately. The text data are evaluated using different TL techniques, including BERT, RoBERTa, DistilBERT and XLNet. Furthermore, the images are analyzed using DL as well as ML techniques to check their efficacy. The best models for both modalities are then chosen for further operations and hybridized to make the final predictions. Therefore, the proposed model is created based on two highly performing techniques to yield the best results. A detailed description of the techniques employed and the hybrid approach proposed is given in the following subsections.

3.4.1.1. *Text classification.* **(i) Bidirectional encoder representations from transformers**

Transfer learning (TL) has recently improved text classification and other computer vision tasks. TL technique makes re-use of the pre-trained model to solve another task. A pre-trained model as the beginning point offers superb benefits despite creating a model from scratch. In this concern, the models based on transformers are being adopted in a broader range by researchers as they do not require labeled data and can be quickly accelerated with the help of GPUs [42]. Therefore, in this paper, one such popular model known as BERT is utilized to perform text classification to detect depression. This model is a pre-trained model that learns the text sequence-wise. Thus, BERT, being bidirectional, processes the text taken as input in both directions compared to other traditional methods that analyze text in only one direction. Due to this property of BERT, high performance can be achieved by capturing many contextual details.

The working of BERT involves embedding input tokens with the help of an embedding layer as the first step, then using a transformer encoder consisting of various self-attention layers to capture distinct parts of the input sequence. Finally, the output of the transformer encoder, contextual token embedding, is given as input to the output layer to make final predictions [43]. The final output generated by the model is the class label, i.e. 0 or 1, representing depressive and non-depressive posts. There are different versions of BERT, but BERT-Base is preferred in this work as it is affordable, smaller and computationally efficient. The basic structure of BERT-Base also explains the architecture of a single encoder, as shown below in Fig. 4, where several encoders are stacked over each other. The used BERT version consists of 12 layers of encoders arranged in a stacking pattern, i.e. 12 attention heads and 110 million parameters. The number of parameters and the attention heads in these models increases with the number of layers. During the training process, masking is performed on the portions of tokens taken as input. Then, the model predicts the masked tokens by learning the association among words and their respective meaning in the context of the input sequence.

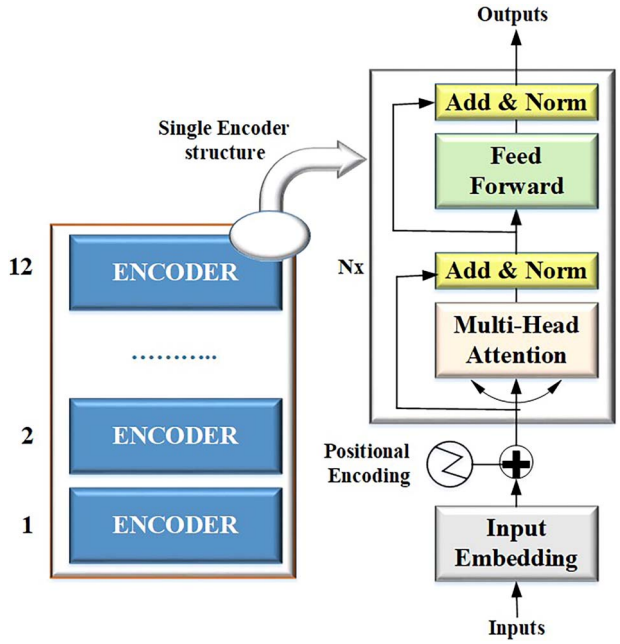

Figure 4. Architecture of BERT-Base Model (Ji et al., 2021).

(ii) Robustly optimized BERT pretraining approach

Robustly optimized BERT pretraining approach (RoBERTa) is another version of the BERT model invented by the AI researchers at Facebook and utilizes a self-attention procedure to evaluate the input sequences. A significant difference between BERT and RoBERTa is that the latter can be efficiently applied to larger datasets and uses an advanced training process. The model can learn generalizations and complex work patterns due to the dynamic-making method used by RoBERTa [44]. Another difference is that RoBERTa uses a long training learning rate and needs to train for longer sequences. RoBERTa neither requires defining tokens belonging to a particular sentence nor using token ids. The architecture of RoBERTa is very much like the BERT model but with some modifications of important hyperparameters.

(iii) DistilBERT

The distilled version of BERT is referred to as DistilBERT. These models are smaller, faster and cheaper, in which the BERT model size is reduced to 40% by knowledge distillation. DistilBERT, with the help of the used distillation procedure, approximates the more extensive neural network with the smaller one [45]. This concept is similar to posterior approximation, which uses the Bayesian statistics theory. This version of BERT does not allow selecting input positions and using token ids like RoBERTa.

(iv) XLNet

XLNet is the newly developed unsupervised language that uses Transformer-XL as the base model and has improved significant performance in textual classification. It utilizes an improvised mechanism for training and possesses enormous computational power compared to BERT. The improvement in the training is made by using modeling referred to as permutation language modeling. The main difference between XLNet and BERT is that in the former, the predictions for all the tokens are performed randomly [46]. In the case of BERT, predictions are only made for those tokens which are masked. Also, the concept behind XLNet contrasts traditional language models where sequential order is followed to make predictions on tokens.

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```
Model: "sequential_3"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d_9 (Conv2D)           (None, 254, 254, 32)      896

 max_pooling2d_9 (MaxPooling  (None, 127, 127, 32)     0
 2D)

 conv2d_10 (Conv2D)          (None, 125, 125, 64)      18496

 max_pooling2d_10 (MaxPoolin  (None, 62, 62, 64)       0
 g2D)

 conv2d_11 (Conv2D)          (None, 60, 60, 64)        36928

 max_pooling2d_11 (MaxPoolin  (None, 30, 30, 64)       0
 g2D)

 flatten_3 (Flatten)         (None, 57600)             0

 dense_6 (Dense)             (None, 128)               7372928

 dropout_3 (Dropout)         (None, 128)               0

 dense_7 (Dense)             (None, 1)                 129

=================================================================
```

Figure 5. Output of model summary.

3.4.2. *Image classification*

(i) DL-based designed CNN

To differentiate users with depression and no depression based on image modality, CNN is used, which is a type of DL technique and is most widely preferred in text analysis. One of the unique properties of CNN is that the parameters, such as batch size, drop out and layers, can be easily varied to achieve the objective function with high accuracy. These models have a layered architecture and are computationally very efficient. Different layers contribute to the building of CNN, such as convolutional, pooling, flattening and fully connected. Also, the output of the model summary is presented in the following Fig. 5.

In this paper, following the model's basic structure, the adjusted CNN is created to perform such classification with high performance, as shown in Fig. 6. Various layers added at different positions in the modeled CNN comprised the following layers [47,48]:

- **Convolutional layer:** The initial extraction of features from the input data is the responsibility of the first layer of CNN, namely the Convolutional Layer. This layer serves as the base for other layers and, therefore, is considered the building block of CNN. This layer makes use of small matrices with size 2×2, 3×3, 5×5 or 7×7 known as filters or kernels, which is convolved with the input image by applying dot product. The output generated from this operation is the feature map given as input to the next layer. In the proposed architecture of CNN, three convolution layers are used at the first, third and fifth positions with different numbers of filters.
- **Max-pooling layer:** A pooling layer is applied to reduce the input size by decreasing its dimensionality. It eliminates the less relevant features from the actual data and makes the entire process computationally fast. This layer is usually inserted between the convolutional layers and only makes improvisations/updations in the required data information. The pooling layer also can solve the overfitting problem and build a new set of pooled feature maps from the old one. In the current model, three such layers are inserted at the second, fourth and sixth positions.
- **Flatten layer:** After the convolutional and max-pooling layer, the flatten layer is added to convert the outputted feature maps into one-dimensional vector form to apply further operations. The result of this layer is then sent to the dense or fully connected layer as input. In the proposed model, one flatten layer is inserted at the seventh position.
- **Dense or fully connected layer:** This comprises the last CNN layer responsible for generating the classified output. The fully connected layer takes the input from the flattened layer and performs mathematical computations to perform accurate classification. The output obtained from this layer is then sent to a logistic function that transforms the outcome into a probability score or label belonging to a particular class. Here, the softmax activation function is adopted to obtain probability distribution from the vector of values and can effectively perform multi-class classification. Two fully connected layers are used at the 8th and 10th positions in the designed network.

$$\text{output} = \text{act}\,(\text{dot}\,(\text{it} + \text{KW}) + \text{bi}) \tag{1}$$

Here, dot represents dot product of weights and input it denotes input data, KW is the weight data and bi denotes biased value.

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- **Dropout layer:** The dropout layer is added at the ninth position in the model architecture and can overcome the overfitting problem. This layer provides stability to the model and makes its processing faster. Due to the addition of this layer, a model can efficiently learn the relevant features of the image. The different parameters are set to perform the classification using designed CNN, such as learning rate = 0.01, dropout = 0.5, momentum = 0.09, batch size = 15 and epochs = 300.**(ii) ML-based models**

(a) KNN: KNN is the widely preferred ML classifier based on a supervised platform where a grouping of data points is performed using proximity. This algorithm assumes the minimum distance between the samples belonging to similar classes. In KNN, firstly, the value of 'K' is set and then a metric is used, such as Euclidian distance, to compute the distance among the 'K' number of neighbors. Based on the value of 'K', the data points with minimum Euclidian distance from classes are selected and the new data point is assigned to the class having the majority of selected data points. KNN is simple to implement and is also known as a lazy learner as it does not use a training dataset for learning immediately but stores the dataset to use during the classification process [49]. The distance calculation between two points using the Euclidean distance (Euc) measure can be done using Equation (2).

$$\text{Euc}\,(u, v) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + . \cdots + (u_s - v_s)^2} \tag{2}$$

Here, u and v are the samples that need to be measured with the help of s characteristics. One of the significant hurdles in KNN is to find out the optimal value of 'K' on which the selection of neighbor depends. Therefore, to avoid ties, it should be chosen in odd numbers.

b) SVM: SVM is the supervised ML algorithm that aims to determine an optimal hyperplane or line that can perfectly classify the data points in N-dimensional space. The hyperplane that provides the maximum distance or margin between the data points of both samples is chosen. Mostly, the hinge loss function is used that performs well in maximizing the margin and can be represented as

$$h\left(u, v, f(u)\right) = \begin{cases} 0 \text{if} v * f(u) \geq 1 \\ 1 - v * f(u), \ \text{else} \end{cases} \tag{3}$$

In SVM, to achieve a low classification error rate, keeping the maximized margin between two classes is mandatory. The role of the kernel in SVM is very significant, which converts the low input space into higher dimensional space efficiently. This kernel trick makes the SVM more accurate, robust and flexible [50].

(c) Decision tree (DT): DT is a tree-structured supervised learning classifier where the dataset features are represented by internal nodes. These trees' branches and leaf nodes indicate a decision rule and the respective outcome. DT starts from the root node and keeps on expanding to branches to perform the operations on the dataset, thus forming a hierarchal tree structure. The matching of the record and root attribute is done until the leaf node of the tree is encountered. Two techniques, namely Gini Index and Information Gain, are mostly preferred to select the root attributes. DT can perform at high speed on large datasets but suffers from the problem of overfitting [51]. Therefore, an ensemble-based technique known as RF offers a solution to the issue in DT by combining multiple DTs [52]. Consider D_n as the data present at node n with s_n samples and t_n thresholds. The algorithm for the classification tree can be presented in Equation (4) as

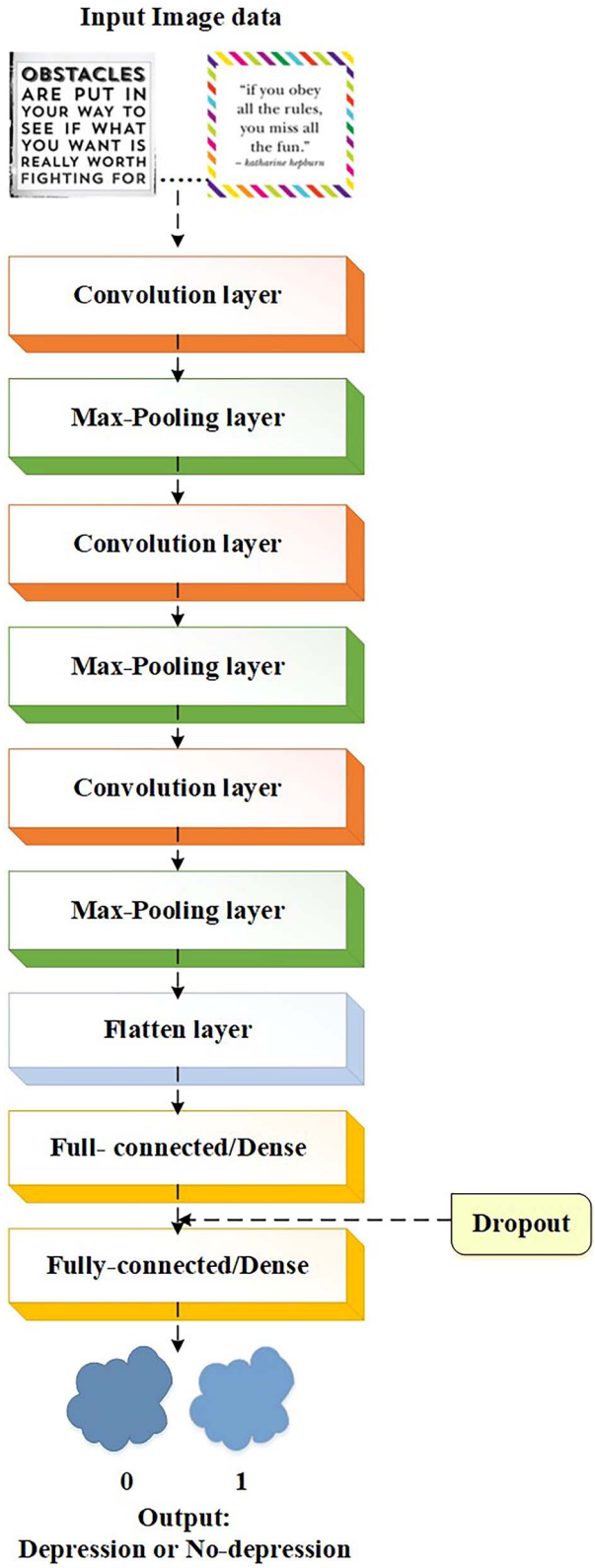

Figure 6. Designed architecture of CNN.

$$G\left(D_n, t_n\right) = \frac{s_n^{\text{Left}}}{s_n} H\left(D_n^{\text{Left}}\left(t_n\right)\right) + \frac{s_n^{\text{Right}}}{s_n} H\left(D_n^{\text{Right}}\left(t_n\right)\right) \tag{4}$$

where H is the left and right impurities measurement at node n and the number of instances is denoted by s_n.

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3.4.3. *Proposed hybrid BERT-CNN approach*

The gaps in the existing literature, including the lack of a multimodal dataset, a robust multimodal model, simultaneous evaluation of text and images, and much less use of hybrid advanced learning platforms, encouraged this study to propose and present a reliable strategy to deal with multimodal data on social media platforms. There are different models that can either work on text or image data for depression analysis but lack the simultaneous evaluation of text and image data. Due to the high variability in the content posted by users on social media, it is necessary to develop a unified model that can recognize whether a person is suffering from depression or not based on multimodal data. Therefore, this study presents a hybrid model combining TL and DL platforms to deal with text and the image simultaneously.

The performance evaluation of all the applied techniques for text and image presented in the previous sections showed the highest performance with BERT and CNN. Therefore, these techniques are selected to create a hybrid approach by combining them to make the final prediction based on prior individual predictions. The text data are classified using BERT, and for images, CNN is adopted due to its significant features and higher accuracy than ML techniques. Afterward, a hybrid model is designed by the combination of these two applied techniques, i.e. BERT+CNN = hybrid approach, for efficiently detecting depressive and non-depressive users. A schematic diagram showing the proposed hybrid BERT-CNN concept, designed to overcome research gaps, is presented in the following Fig. 7.

In depression detection, textual data have been given more preference among all the modalities due to their more accurate results. Therefore, in this work, the separate predictions made by BERT and CNN on text and image data are taken as input in the ratio of 70:30, i.e. 70% text and 30% images. This study considered several combinations of this text:image ratio, such as 50:50, 60:40 and 80:20; however, the best results are achieved with the abovementioned 70:30 ratio. As shown in Fig. 7, the collected data are first preprocessed using the procedures and techniques elaborated in previous sections. The preprocessed text data with corresponding images are then fed separately as input to BERT and CNN models. The BERT model comprises certain transformer encoder layers built to process the text data and generate the output by applying certain mathematical operations. Similarly, the image data are passed through several CNN layers by increasing the number of abstractions to capture useful information about the image features. The fully connected layer of CNN generates the output, classifying the image into depressive or non-depressive. A thorough structure of the designed BERT and CNN is given in Sections 3.4.1 and 3.4.2. Finally, the outcomes of both models are fused using an aggregation technique known as Soft voting classifier to generate the final class prediction or label. Based on the probability of predictions, this voting procedure works to hybridize the predictions from BERT and CNN, thus forming a multimodal text and image model. In Soft voting classifier algorithm, each model assigns a probability value to each class, indicating that a specific data point refers to a particular class. In our case, there were two classes, i.e. 0 and 1, representing depressed and non-depressed posts. The prediction probabilities by the models for each class are summed up and the final prediction is simply the class yielding the highest probability value. Therefore, following this procedure, the hybrid model is designed and the final results are generated to detect the depression accurately.

3.5. Performance evaluation

To build an effective ML model, it is necessary to evaluate the performance of the model. The quality of the model can be analyzed using various performance evaluation metrics, which tell how well the model performs on the given data. Based on the results obtained, the hyperparameters of the model can be adjusted to enhance its performance. In this paper, metrics such as accuracy, F1-score, recall and precision are utilized to measure the performance and robustness of the proposed model with other techniques. These metrics evaluate the performance based on the model's output in the form of class labels, such as 0 or 1, representing depressive and non-depressive users. Four main parameters, including true positive (TP), true negative (TN), false positive (FP) and false negative (FN), are used in the computation of these metrics having different meanings. TP correctly indicates the presence of a state and TN correctly indicates the absence of a state. Similarly, FP denotes the wrong indication of the presence of a state and TN is the wrong indication of absence of a state. Mathematically, the evaluation metrics can be represented as [53].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{5}$$

$$\text{Sensitivity/Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{F1} - \text{score} = \frac{2}{\frac{1}{\text{recall}} + \frac{1}{\text{precision}}} \tag{8}$$

Accuracy is the ratio of the number of accurate predictions to the total number of predictions. Recall or sensitivity represents the number of TP outputted by the model. Precision indicates the ability of the model to identify a specific class. F1-Score is the harmonic mean of precision and recall. The values of all these metrics lie between 0 and 1.

4. EXPERIMENTAL RESULTS AND ANALYSIS

The entire training process was performed on Anaconda Navigator in Python programming platform with GPU system using the TensorFlow deep learning tool and Keras library. The experimentation was performed for three models adopted in this work: (i) BERT for textual data, (ii) CNN model for image data and (iii) a hybrid approach combining BERT-CNN for multimodal data. Each of the models is evaluated one by one and their respective comparison is also performed with other TL and ML techniques to find the best one. To deal with text data, BERT is applied, which yielded an accuracy of 97%, sensitivity or recall of 97%, precision of 97% and F1-score of 97%, respectively. Furthermore, the BERT model is analyzed by comparing it with other versions of BERT, including RoBERTa, DistilBERT and XLNet. The results obtained reveal 81.20% of accuracy, 75% of sensitivity, 74% of precision and 74% of F1-score with RoBERTa; 81.50% of accuracy, 76% of sensitivity, 76% of precision and 76% of F1-score with DistilBERT; and 12.72% of accuracy, 36% of sensitivity, 54% of precision and 25% of F1-score with XLNet. The overall comparison based on evaluation metrics demonstrates the highest performance with the BERT model compared to other TL models in the classification of textual data. The result values obtained for all the models

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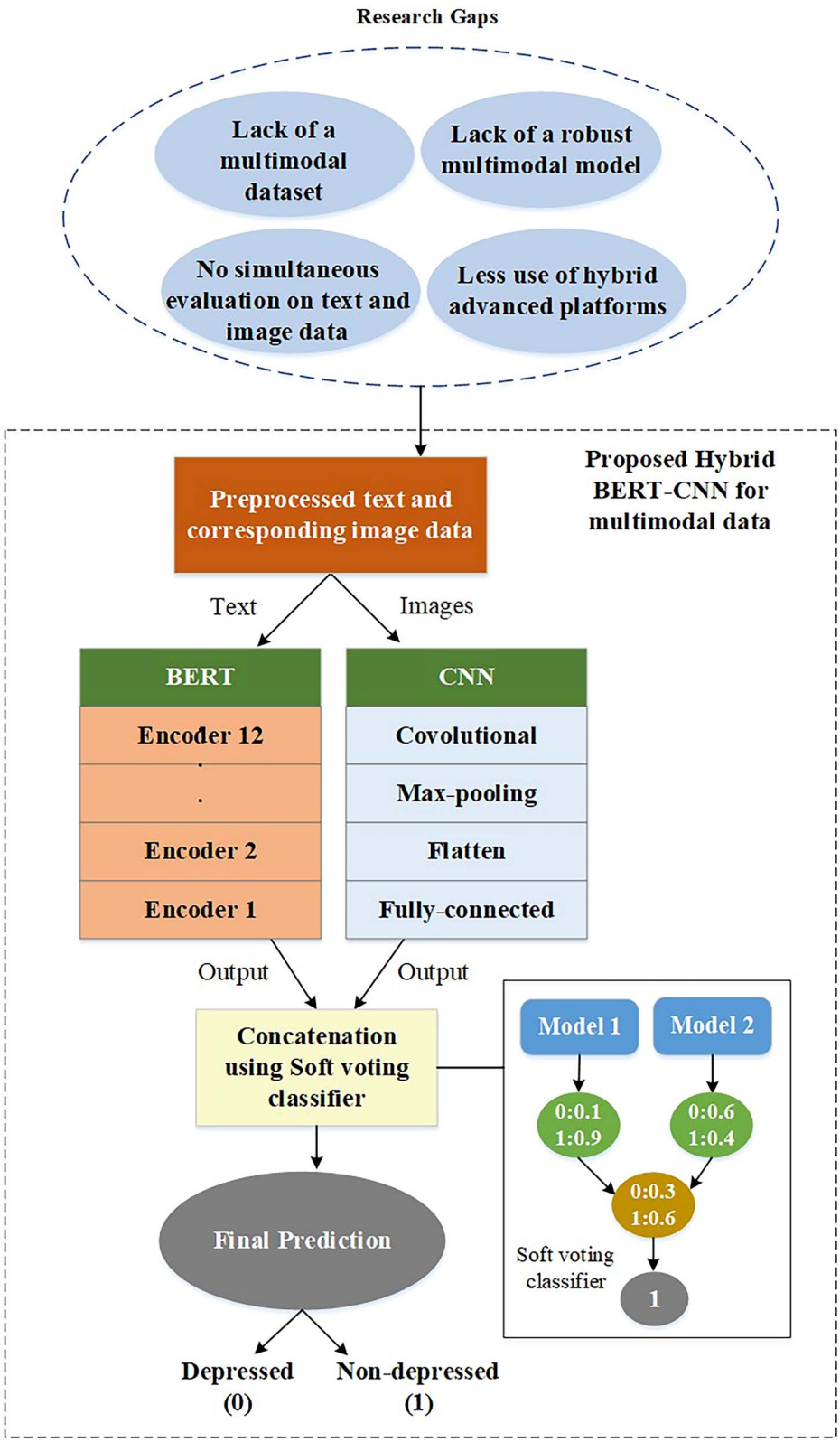

Figure 7. Concept of hybrid BERT-CNN.

employed for textual data are provided in the following Table 3 and are graphically shown in Fig. 8.

After the evaluation of textual posts, the classification of image data is performed with the help of the designed CNN model due to its tremendous properties in image-related tasks. The results achieved using the applied DL-based CNN model reflect the significant performance by attaining an accuracy of 89%, sensitivity or recall of 83%, precision of 86% and F1-score of 83%, respectively. The experimentation is extended by considering the ML models in this process, where KNN, SVM and DT were

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Table 3. Results for classification of text data into depressive and non-depressive.

Model	Accuracy	Sensitivity/recall	Precision	F1-score
BERT	97%	97%	97%	97%
RoBERTa	81.20%	75%	74%	74%
DistilBERT	81.50%	76%	76%	76%
XLNet	12.72%	36%	54%	25%

Table 4. Results for classification of image data into depressive and non-depressive.

Model	Accuracy	Sensitivity/recall	Precision	F1-score
CNN	89%	83%	86%	83%
KNN	73%	73%	79%	73%
DT	81%	81%	82%	81%
SVM	81%	81%	82%	81%

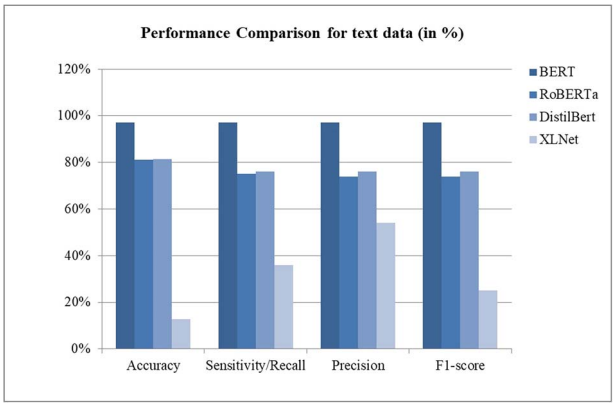

Figure 8. Graph showing results for text data classification using different techniques.

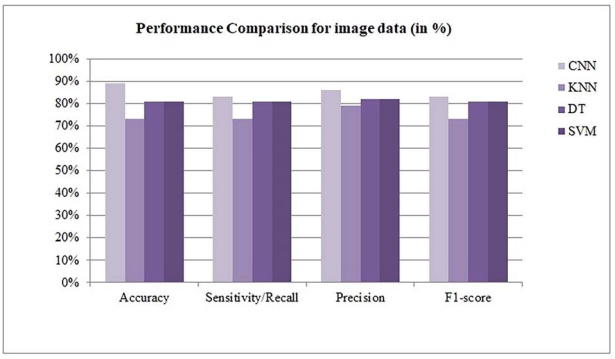

Figure 9. Graph showing results for image data classification using different techniques.

employed to check their efficacy. The accuracy rates of 73%, 81% and 81%; the sensitivity of 73%, 81% and 81%; the precision of 79%, 82% and 82%; and the F1-score of 73%, 81% and 81% were obtained using ML techniques, i.e. KNN, DT and SVM. The comparative analysis of results obtained using DL and ML models shows that the efficiency of the CNN model outperforms the ML techniques by achieving the highest accuracy rate. The result values for different parameters attained for CNN and ML methods are given in Table 4 and are graphically presented in Fig. 9.

The graph in Figs 8 and 9 reveals the highest performance using BERT and CNN for textual and image data. Therefore, to take advantage of both techniques in this study, a hybrid approach is proposed by combining BERT and CNN, i.e. BERT-CNN. To evaluate the proposed BERT-CNN model, the final prediction is made with 70% weightage given to textual data while the remaining 30% weightage is given to image data due to more authenticity of the textual information. The results achieved using the proposed hybrid approach are shown in Table 5 and reveal the best performance. Furthermore, to validate the results, the proposed BERT-CNN is comparatively evaluated with other combinations such as BERT-KNN, BERT-SVM and BERT-DT. The performance evaluation metrics are computed for each model and the results achieved proved the efficacy of the proposed approach by yielding an accuracy rate of 99% in comparison to other methods with accuracy of 58.30%, 58.20% and 65.10%, respectively, for BERT-SVM, BERT-KNN and BERT-DT.

Figure 10 shows that the highest performance values are attained by applying the proposed technique, and BERT-KNN shows the lowest results. Therefore, this study evaluated each algorithm separately and then in combination to cross-validate all the results. Firstly, the text classification techniques are analyzed by evaluating them individually. Among all, BERT has shown the highest accuracy for text classification into depressive and non-depressive categories. Afterward, a DL-based CNN method is evaluated individually in comparison with four ML algorithms, out of which CNN has outperformed the other ML methods in detecting depressive and non-depressive users from images. Finally, the best-performing models, i.e. BERT and CNN, hybridized to make a unified and robust model (BERT-CNN) that can work with every type of data efficiently.

Furthermore, the comparison of the proposed hybrid approach attained the highest results compared to other hybrid models. The results obtained in the above tables demonstrate the efficiency of CNN in extracting the most relevant features from the data by increasing the level of abstraction and adjusting the model's parameters. Also, the unique feature of the BERT model to process data in bidirectional allows for extracting more contextual details and improving the model's overall performance. The increase in accuracy rate is also observed when BERT and CNN are used in combination to classify multimodal data to detect depression. Therefore, the best accuracy is achieved when deep feature extraction capabilities of the CNN model are

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Table 5. Results for classification of combined data into depressive and non-depressive using the proposed method.

Model	Accuracy	Sensitivity/recall	Precision	F1-score
BERT-CNN (proposed)	99%	97%	100%	99%
BERT-SVM	58.30%	51%	56%	52%
BERT-KNN	58.20%	51%	56%	52%
BERT-DT	65.10%	58%	63%	59%

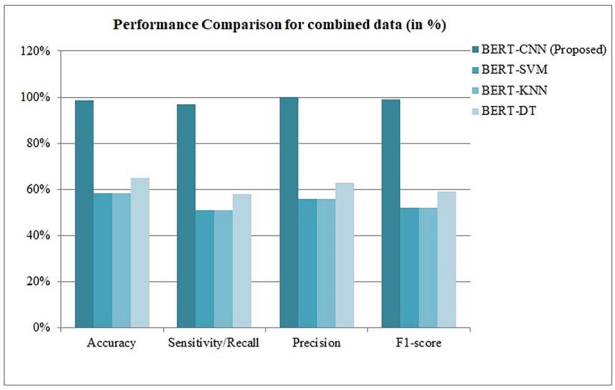

Figure 10. Graphical analysis of results using the proposed method on multimodal data.

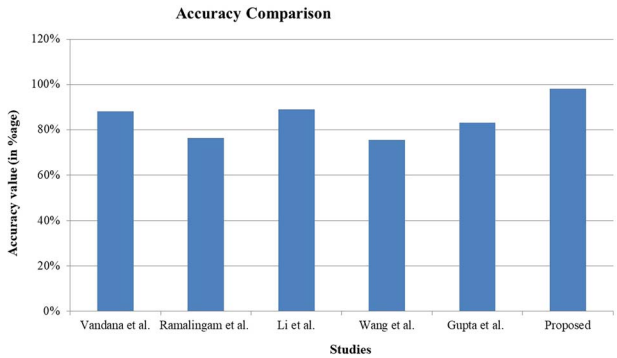

Figure 11. Comparison analysis of the proposed approach with state-of-the-art based on the accuracy.

combined with the BERT model for an efficient image and text classification.

4.1. Statistical analysis

The results achieved in this research showed the highest performance with the proposed approach, which is then validated by using the significantly known statistical test named as Friedman's Rank Test (FRT). To perform statistical analysis using FRT in order to evaluate performance of different models and the proposed approach, two other datasets, i.e. [54,55], are also considered. Both datasets contain depression-related data, which is the area of focus of this research study. The different accuracy rates achieved by the employed and the proposed approaches on the considered datasets are given in Table 6 below, upon which the FRT is performed. The FRT helps to validate the substantial differences between the models. An evaluation is performed based on ranks after forming the null hypothesis. Consequently, a null hypothesis states that all the methods exhibit the same accuracy rate and no considerable visible difference. Moreover, the FRT allocates the highest rank to the model with the highest accuracy rate and the lowest to the model with the lowest accuracy rate. The test results attained a P-value of 0.0421, lower than the significance level, i.e. $P < 0.05$. The test results showcased the fact that there is a significant difference in model performance on different datasets, subsequently rejecting the null hypothesis. The highest rank, i.e. rank 4, is attained by the proposed model, representing the highest performance and the lowest rank is attained by the BERT_SVM model. Table 7 below shows the FRT results, providing the average rank of methods used and the respective P-value.

4.2. Comparison with state-of-the-art studies

The proposed approach is compared with the current state-of-the-art studies to analyze the robustness and efficacy. Accuracy is utilized as the performance evaluation measure, which has been employed due to its high adoption in all the research works for comparison purposes. Table 7 provides an insight into this comparison that can be visualized graphically in the following Fig. 11. Ramalingham et al. [56] used Twitter data in the form of images and videos to analyze depression with the help of SVM. They achieved accuracy rates of 82.2% and 70.5% for predicting males and females with depression.

Similarly, another study by Wang et al. [21] employed BERT, SVM, NB, LR, CNN and LSTM on text, tags and emoticons data acquired from the Weibo platform. The highest accuracy rate of 75.6% was attained by using the BERT model, thus showing its efficacy in depression prediction. The analysis of depression using electroencephalography (EEG) signal data was performed by Li et al. [57]. The EEG data were fed as input to SVM, CNN, an ensemble of deep forest, RF and KNN, and the highest results were obtained with LSTM, i.e. 83%. The present study considered evaluating multimodal data (text, emoticons and images), which previous studies have not utilized much. The study aims to propose a robust approach that can be used for all types of data to detect users with depression and no depression. The data in Table 8 indicate that the proposed hybrid BERT-CNN approach has outperformed the current state-of-the-art with an accuracy of 99%. Therefore, the proposed model has a high potential to predict depression with optimum performance.

5. CONCLUSION AND FUTURE WORK

Social media platforms have become the most widely used source for users to share their feelings, emotions, thoughts, views, etc., with their friends and family through pictures, text, audio, videos, etc. Analyzing such data in-depth can provide crucial clues regarding a person's state of mind. Among mental disorders, depression seems to be a rapidly growing disease, particularly in the younger generation worldwide. Therefore, the early detection of depression is of utmost importance nowadays to save people's lives by providing them with timely counseling and treatment.

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Table 6. Accuracy achieved by models on different datasets.

Datasets/models	Proposed	BERT_SVM	BERT_KNN	BERT_DT
Ours	0.9900	0.5830	0.5820	0.6510
[54]	0.9030	0.6250	0.6740	0.7330
[55]	0.9280	0.6690	0.6920	0.7540

Table 7. Results of FRT

Rank	1st	2nd	3rd	4th	P-value
Model	BERT_SVM	BERT_KNN	BERT_DT	Proposed	0.0421
Average rank	1.3333	1.6667	3	4	

Table 8. Comparison of the proposed approach with previous studies

Author	Year	Modality	Classifier	Dataset	Average accuracy
Vandana et al.	2023	Text and audio data	Textual CNN, audio CNN, LSTM and Bi-LSTM	DAIC-WoZ	88% (with Bi-LSTM)
Ramalingam et al.	2019	Twitter-like data (videos, pictures)	SVM	Weibo posts	82.2% (for males) 70.5% (for females)
Li et al.	2019	HydroCel Geodesic Sensor Net	SVM, CNN, Ensemble of deep forest and SVM, RF, KNN	EEG features	Highest with ensemble = 89.02%
Wang et al.	2020	Text, tags and emoticons	BERT, SVM, NB, LR, CNN and LSTM	Weibo	Highest with BERT = 75.6%
Gupta et al.	2020	Text	SVM, DT, KNN, LR and LSTM	Kaggle and Twitter	Highest with LSTM = 83%
Proposed study		Text, images and emoticons	BERT (text), CNN (images), SVM, hybrid BERT-CNN (proposed)	Instagram and other social media posts	Highest with proposed BERT-CNN = 99%

This study aims to provide a robust multimodal strategy to detect depression using a hybrid approach. In this study, three models are designed: firstly, BERT for text; secondly, CNN for images; and thirdly, hybridization of BERT and CNN model for multimodal data, i.e. text + images. The current state-of-the-art analysis reveals very little use of BERT and CNN for text and image data and also attained mediocre performance in depression detection. Therefore, this study employs BERT and CNN models individually and combined to achieve higher accuracy.

Moreover, a dataset has been created consisting of posts from Instagram to classify depressive and non-depressive users. For text data, BERT and other versions of BERT, namely RoBERTa, DistilBERT and XLNet, are applied and the experimental results show the best accuracy with BERT, i.e. 97%. Similarly, for image data, CNN and ML models such as SVM, KNN and DT are employed and among all, CNN has achieved the best accuracy with 89%, thus proving the efficacy of the deep learning CNN model in this research. A robust model that can work on both text and image data is proposed by combining BERT and CNN. The proposed hybrid approach is compared with other combinations, including BERT-SVM, BERT-KNN and BERT-DT. The results showed that the proposed hybrid approach BER-CNN has achieved the highest accuracy rate, i.e. 99%. Lastly, based on the accuracy parameter, it is found that the proposed approach has outperformed the current state-of-the-art studies. Therefore, it is concluded that the best results can be attained by combining the CNN model to extract deep features and the BERT model for multimodal data classification into depressive and non-depressive posts. Furthermore, the proposed strategy can be helpful for health practitioners and researchers to perform efficient sentiment analysis.

However, the proposed work yielded satisfactory results, but certain aspects need to be focused upon in the future. Firstly, audio and video modality, along with text, emoticons and image data, can be considered for better assessment. It may enhance the performance of the model in depression detection. Secondly, a hybrid approach combining text, image, audio and video features can be designed for a more reliable prediction. An ensemble approach combining multiple models can be utilized to produce improved results. Thirdly, the experiment can be extended by evaluating multiple benchmark datasets related to depression. The data samples can be increased by acquiring the data from various social networking platforms despite focusing on only one or two. Lastly, the person should be analyzed at different severity levels, such as early, moderate and severe also, to enable the treatment of depression at an early stage.

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DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

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APPENDIX

Acronym	Meaning	Acronym	Meaning
WHO	World Health Organization	**MSMTC**	Medical social media text classification
DALYs	Disability-adjusted life years	**RNN**	Recurrent neural network
AI	Artificial intelligence	**LR**	Linear regression
ML	Machine learning	**DCNN**	Deep convolutional neural network
NLP	Natural language processing	**HI**	Happiness Index
SVM	Support vector machine	**OCR**	Optical character recognition
LSTM	Long short-term memory	**RoBERTa**	Robustly optimized BERT pretraining approach
TL	Transfer learning	**TP**	True positive
DT	Decision trees	**TN**	True negative
DL	Deep learning	**FP**	False positive
CNN	Convolutional neural network	**FN**	False negative
BERT	Bidirectional encoder representations from transformers	**NB**	Naïve Bayes
KNN	K-Nearest neighbor	**MNB**	Multimodal naïve Bayes
RF	Random forest	**TAN**	Transductive adversarial network
BiGRU	Bidirectional rated recurrent units	**BF**	Bayes factor
EEG	Electroencephalography	**FRT**	Friedman's Rank Test

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